

Debopam Goswami

📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX, 77843

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🔍 RESEARCH SUMMARY

My research in sub-GeV dark matter (DM) and neutrino phenomenology spans both lab-based and astrophysical searches. I have developed analytical and computational tools for multi-body scattering that have improved sensitivity across a range of neutrino and DM experiments for several dark-sector models. In my ongoing work, I am applying machine learning techniques to enhance signal discrimination. More broadly, I am interested in leveraging artificial intelligence to automate modeling, simulation, and scientific discovery within particle physics and related disciplines.

🎓 EDUCATION

Texas A&M University (TAMU), College Station

Doctor of Philosophy (Ph.D.) in Physics

Advisor: Prof. Bhaskar Dutta | GPA: 4.0/4.0

Texas, USA

Aug 2021 – Present

Indian Institute of Technology (IIT) Kanpur

Master of Science (M.Sc.) in Physics

Thesis: Dark Matter Genesis in the Early Universe

Advisor: Prof. Debtoosh Chowdhury | CGPA: 9.3/10.0 | **Rank: 1**

Uttar Pradesh, India

July 2019 – May 2021

University of Calcutta (CU)

Bachelor of Science (B.Sc.) in Physics

Aggregate: 78.5% | First Class with Distinction

West Bengal, India

July 2016 – May 2019

📖 PUBLICATIONS

   

1. B. Dutta, **D. Goswami**, B. Hu, W. C. Huang, V. Pandey, *Dark Matter-Induced Nuclear De-Excitation at SBND with Ab Initio Nuclear Theory*, [[arXiv:2602.02817](#)]
2. B. Dutta, **D. Goswami**, A. Karthikeyan, *Monophotons from Scalar Portal Dark Matter at Neutrino Experiments*, *JHEP* 033P 0825 (2026) [[arXiv:2507.07936](#)]
3. B. Dutta, **D. Goswami**, A. Karthikeyan, K. J. Kelly, *Dirt/Detector/Dump: Complementary BSM production at Short-Baseline Neutrino Facilities*, *J. High Energ. Phys.* 2025, 240 (2025) [[arXiv:2501.09840](#)]
4. S. Akhtar, S. Choudhury, S. Chowdhury, **D. Goswami**, S. Panda, A. Swain, *Open Quantum Entanglement: A study of two atomic system in static patch of de Sitter space*, *Eur. Phys. J. C* 80, 748 (2020) [[arXiv:1908.09929](#)]

SOON TO APPEAR...

5. B. Dutta, **D. Goswami**, A. Karthikeyan, V. Pandey, Z. Tabrizi, A. Thompson, R. G. Van de Water, *Enhancing New Physics Searches with a BNB Beam Dump Configuration*, [[arXiv:2602.*****](#)]
6. C. Cappiello, P.S.B. Dev, J.B. Dent, B. Dutta, **D. Goswami**, A. Karthikeyan, *Boosted dark matter from single, stacked blazars and cosmic protons*, [[arXiv:2602.*****](#)]
7. B. Dutta, **D. Goswami**, J. Kumar, M. Rai, D. Sathyan, *Explanation of Galactic Center GeV Excess via Cosmic Ray-Dark Matter Scattering*, [[arXiv:2602.*****](#)]

1. Graduate Researcher

Aug 2021 – Present

Advisor: Prof. Bhaskar Dutta | Mitchell Institute for Fundamental Physics & Astronomy

Texas, USA

Accelerator-Based Dark Matter & HNL Searches

- Investigated sub-GeV heavy neutral leptons (HNLs) with transition magnetic moments in the Fermilab Short-Baseline Neutrino (SBN) program, emphasizing beam-dump production and exploiting high-Z enhancement in iron dumps.
- Identified distinctive kinematic signatures from production in dump, dirt, and detector regions to improve signal-to-background discrimination.
- Improved SBND sensitivity to light dark matter via localized low-energy electromagnetic activity ("blips") using state-of-the-art ab initio nuclear structure calculations including argon excited states up to 18 MeV.
- Analyzed monophoton signatures from $2 \rightarrow 3$ dark matter-nucleus scattering processes, mediated by scalar particle, in DUNE ND, SBN program, and CCM200, incorporating detailed energy, angular, timing, and spatial distributions.

Astrophysical & Boosted Dark Matter Phenomenology

- Proposed a novel mechanism for the explanation of Galactic Center gamma-ray excess using cosmic-ray proton scattering off inelastic dark matter or elastic dark matter with 2 mediators in the Milky Way halo.
- Demonstrated excellent fit with observed Galactic Center gamma-ray excess within allowed parameter space.
- Derived improved constraints on dark photon and light scalar mediator models using blazar-boosted dark matter scenarios.
- Performed comprehensive sensitivity analyses across 10 experiments (DUNE FD, IceCube, Hyper-K, JUNO, LZ, PandaX-4T, Borexino, etc.), including both single and stacked blazar analyses.

Theoretical & Computational Developments

- Developed scalable Python packages implementing analytical expressions for multi-body ($2 \rightarrow 3$ and $2 \rightarrow 4$) scattering processes relevant in neutrino, dark matter experiments, and blazar/cosmic boosted dark matter.
- Developed Python-based computational frameworks for different kinematic spectra modeling, and sensitivity projections across multiple detectors over various BSM models.
- Automated BIGSTICK implementation for computing nuclear transition strengths across isotopes.

Ongoing & Future Work

- Applied machine learning techniques to the SBN program at Fermilab and various astrophysical sources.

2. Dark Matter Genesis in the Early Universe.

Jan 2020 - May 2021

Advisor: Prof. Debtosh Chowdhury | Indian Institute of Technology (IIT) Kanpur

Uttar Pradesh, India

- Investigated the mechanisms responsible for non-thermal dark matter production in the early universe.
- Explored axion as a viable dark matter candidate, analyzing constraints on its mass, coupling strength, and natural emergence in theories beyond the Standard Model.
- Performed quantitative calculations on relic photon and neutrino abundances, WIMP miracle, dark matter density via the Lee-Weinberg bound, and solutions of the Boltzmann equation.

3. Perturbative & Non-Perturbative Physics and Instantons.

Jun 2019 - July 2019

Advisor: Prof. Sayantan Sharma | The Institute of Mathematical Sciences (IMSc)

Tamil Nadu, India

- Calculated vacuum functionals for harmonic and anharmonic oscillators using semi-classical path integral methods. Investigated asymptotic expansions of integrals, asymptotic solutions to differential equations, summation of divergent series, and instanton concepts in quantum mechanics.

- Determined ground state corrections for the anharmonic quartic oscillator using perturbation theory and applied asymptotic relations to the double well potential. Used Borel re-summation technique to extract meaningful results from non-convergent energy perturbation series.

4. Path Integral Technique in Quantum Field Theory.

Nov 2019 - Dec 2019

Advisor: Prof. Ashoke Sen | Harish-Chandra Research Institute (HRI)

Uttar Pradesh, India

- Developed expertise in the Feynman Path Integral technique in quantum field theory and studied Feynman diagrams. Applied these methods to Klein-Gordon, interacting scalar, complex scalar, and Dirac field theories.
- Worked on the fundamentals of quantum electrodynamics and calculated cross sections for various scattering processes.

5. Linear Algebra and Group Theory.

Sep 2017 - Nov 2017

Advisor: Prof. Sourov Roy | Indian Association for the Cultivation of Science (IACS)

West Bengal, India

- Developed expertise in group theory with a focus on the properties of S_3 , normal subgroups, quotient groups, homomorphisms, isomorphisms, automorphisms, conjugacy classes, and permutation groups.

TALKS AND PRESENTATIONS

1. **PROSPECT Experiment** : (Online) Jul 2025
Talk: *Probes of BSM Models in PROSPECT Experiment.*
2. **Phenomenology Symposium** : University of Pittsburgh May 2025
Talk: *From Neutron Stars to Beam Dumps: 2-to-3 Process with Single Photon Final State.* [\[Link\]](#)
3. **Particle Physics on the Plains** : University of Kansas Nov 2024
Talk: *Signals from Cosmic Boosted Strongly Interacting Dark Matter.* [\[Link\]](#)
4. **APS DPF-Phenomenology Symposium** : University of Pittsburgh May 2024
Talk: *Utilizing the Iron Dump at SBN Facilities to Probe Heavy Neutral Lepton and Dark Matter.* [\[Link\]](#)
5. **Seminar Speaker** : IIT Kanpur April 2021
Talk: *Dark Matter Genesis in the Early Universe.*

CONFERENCES AND WORKSHOPS

(Excluding the conferences where I have delivered a talk or presentation).

1. **The Mitchell Conference on Collider, Dark Matter, & Neutrino Physics.** May 2024
Texas A&M University
2. **The Mitchell Conference on Collider, Dark Matter, & Neutrino Physics.** May 2023
Texas A&M University
3. **Physics of the Early Universe (Online)** Aug 2020
The International Centre for Theoretical Sciences
4. **XVIIITH Annual Basic Sciences Summer School.** May 2016
Indian Association for the Cultivation of Science (IACS)

TECHNICAL SKILLS

- **Programming Languages** – Python, C++, Bash/Shell, SQL, Wolfram Mathematica, R.
- **Libraries** – NumPy, Pandas, SciPy, JAX, Matplotlib, Seaborn, TensorFlow, PyTorch, LangChain, XGBoost, Scikit-learn, Multiprocessing, Git/GitHub.
- **Machine Learning** – Gradient Boosted Trees, Logistic Regression, Random Forest, Multi-Layer Perceptron, Large Language Models, Retrieval-Augmented Generation.
- **Tools** – BIGSTICK, MadGraph, LaTeX, High Performance Computing, Monte Carlo, Jupyter, Microsoft Office.
- **Operating Systems:** macOS, Linux (Ubuntu), Windows.
- **Soft Skills** – Problem solving, critical and independent thinking, attention to detail, teamwork.

ADDITIONAL PROJECTS

- **Large Language Model (LLM) Powered Scientific Paper Summarizer** Nov 2025 - Present
Built a Retrieval-Augmented Generation (RAG) system using Meta's FAISS and Gemini LLM to summarize 2.4M+ arXiv scientific papers with sub-second query response times, and implemented a Streamlit UI for interactive querying (Work in Progress...).

AWARDS AND HONORS

1. Awarded the **Graduate Student Travel Grant twice** (2024) by Texas A&M University, for participation in conferences, recognizing exceptional academic achievement and research contributions. [\[Link\]](#)
2. Awarded the **General Proficiency Medal 2021** for achieving the **best academic performance** among graduating M.Sc. students in the Department of Physics at IIT Kanpur, India. [\[Link\]](#)
3. Received the **Academic Excellence Award** for two consecutive academic years (2019–2020 and 2020–2021) at IIT Kanpur, acknowledging consistent academic distinction and top-tier performance. [\[Link\]](#)
4. Secured **All India Rank 10** (99.9th percentile) among 14,000+ candidates in Joint Entrance Screening Test (JEST) 2019, the National screening test for Ph.D. programs. [\[Link\]](#)
5. Secured **All India Rank 59** (99.6th percentile) among 16,000+ candidates in IIT-Joint Admission Test (IIT-JAM) 2019, the National screening test for M.Sc. programs in IITs. [\[Link\]](#)
6. Awarded the **Prof. Anil Sen Memorial Prize** by University of Calcutta for achieving the highest score in Physics Practical Examination in B.Sc. 1st, 2nd & 3rd-year exams taken together. [\[Link\]](#)
7. Ranked in **All India Top 20** (99.8th percentile) among 13,000+ candidates in the National Graduate Physics Exam (NGPE), conducted by Indian Association of Physics Teachers (IAPT). [\[Link\]](#)

TEACHING EXPERIENCE

- PHYS 206 University Physics: Newtonian Mechanics for Engineering and Science Fall 2024, Spring 2022
- PHYS 206 Don't Panic: Newtonian Mechanics for Engineering and Science Spring 2024
- PHYS 227 Electricity and Magnetism Laboratory for the Science Fall 2023, Spring 2023
- PHYS 226 Physics of Forces and Motion Laboratory Spring 2023, Fall 2022
- PHYS 207 Don't Panic: Electricity and Magnetism for Engineering and Science Summer 2022
- PHYS 207 University Physics: Electricity and Magnetism for Engineers and Scientists Fall 2021

✓ CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** - DeepLearning.AI : [\[Certificate\]](#) Jul 2023
- **Python Data Structures** - University of Michigan : [\[Certificate\]](#) Dec 2023

📁 LEADERSHIP & ENGAGEMENT

1. **Texas A&M Graduate Consulting Club (TAMUGC)** [\[Link\]](#) College Station, Texas
Pro Bono Project: AI in Action: Redefining Talent Management for a New Era of HR Feb 2025 - June 2025
 - Led a pro bono consulting project focused on AI-enabled hiring systems, authored a research article, developed a 30-item best practices toolkit, and designed a two-day client workshop.
 - Presented deliverables to senior stakeholders, and refined content based on feedback.
2. **Texas A&M University (TAMU)** [\[Link\]](#) College Station, Texas
Graduate Teaching Assistant Aug 2021 - Dec 2024
 - Supervised 9 courses for over 3 years, mentoring and grading 500+ undergraduate students.
 - Developed strong communication skills by teaching complex physics concepts to diverse audiences with instructional time of 300+ hours.
3. **Indian Graduate Student Association (IGSA)** [\[Link\]](#) College Station, Texas
Director, Editorial Team July 2024 - Oct 2024
 - Managed social media platforms in collaboration with the editorial team to enhance outreach and engagement.
 - Contributed to the design and development of planners for events, ensuring effective organization and execution.

👥 REFERENCES

1. **Prof. Bhaskar Dutta**
 - 📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, USA
 - ✉ dutta@physics.tamu.edu
 - 🌐 [Bhaskar Dutta Website](#)
2. **Prof. Bhupal Dev**
 - 📍 Department of Physics, Washington University in St. Louis, USA
 - ✉ bdev@wustl.edu
 - 🌐 <https://sites.wustl.edu/bdev/>